



Cite this article: Rouleau N, Levin M. 2026
Brains and where else? Mapping theories of
consciousness to unconventional
embodiments. *Phil. Trans. R. Soc. A* **384**:
20250082.

<https://doi.org/10.1098/rsta.2025.0082>

Received: 25 April 2025

Accepted: 17 September 2025

One contribution of 18 to a theme issue ‘World
models in natural and artificial intelligence’.

Subject Areas:

artificial intelligence, robotics, software,
systems theory

Keywords:

consciousness, aneural cognition,
functionalism, organizational invariance,
unconventional minds

Authors for correspondence:

Nicolas Rouleau

e-mail: nrouleau@wlu.ca

Michael Levin

e-mail: michael.levin@tufts.edu

Brains and where else? Mapping theories of consciousness to unconventional embodiments

Nicolas Rouleau^{1,2} and Michael Levin^{2,3}

¹Department of Health Sciences, Wilfrid Laurier University, Waterloo, Ontario, Canada

²Allen Discovery Center, Tufts University, Medford, MA, USA

³Wyss Institute for Biologically Inspired Engineering, Harvard University, Boston, MA, USA

NR, 0000-0003-4592-6280; ML, 0000-0001-7292-8084

It is assumed that a useful theory of consciousness (ToC) will explain why consciousness is associated with brains. However, the findings of evolutionary biology, developmental bioelectricity and synthetic bioengineering reveal ancient pre-neural roots of many mechanisms and algorithms occurring in brains: minds may have preceded brains. Most work in the emerging field of diverse intelligence emphasizes externally observable problem-solving competencies in unconventional media, such as cells, tissues and life-technology chimeras. Here, we inquire about the implications of these developments for ToCs. Specifically, we analyse popular current ToCs to ask: What features of each theory specifically pick out brains as a privileged substrate of inner perspective, or do the features emphasized by the theory occur elsewhere? We find that the operations and functional principles of most ToCs are not confined to neural substrates, and that the focus on brains is more driven by convention than by the specific content of existing ToCs. Encouragingly, several contemporary theorists have made explicit efforts to apply their theories to synthetic systems in light of recent technological developments in artificial intelligence and organoid bioengineering. We suggest that the science of consciousness should remain open to minds in unconventional embodiments.

This article is part of the theme issue ‘World models in natural and artificial intelligence’.

© 2026 The Authors. Published by the Royal Society under the terms of the Creative Commons Attribution License <http://creativecommons.org/licenses/by/4.0/>, which permits unrestricted use, provided the original author and source are credited.

1. Introduction

What processes or algorithms underlie the ability of certain physical objects, such as living bodies, to form memories, implement decision-making, have an inner perspective with preferences and navigate their environment in a goal-directed manner? This question is crucial to our understanding of ourselves, the development of ethics and social systems and the status of hybrid and fully synthetic artificial intelligence (AI). A number of formalisms, including that of Turing [1] and later, the connectionist paradigm [2–7], have attempted to delineate what it is that confers mind within a given embodiment. In all cases, however, it was assumed that the formalisms' dynamics describe events in brains. What would have happened if, instead of touting the functional capacities of brains, the biologists came to pioneers in the field of AI, such as McCulloch [3], Pitts, Papert, Rosenblatt, etc. and said: 'we were wrong, the human mind is in the liver'? Would they have had to modify their models, or shrug it off as irrelevant to their efforts? What is actually 'neural' about artificial neural networks, used to attempt to understand minds and build AI? Specifically, what aspects of neuroscience-inspired formalisms are actually tied to neurons *per se*, as opposed to other cell types?

This seems like an odd question—surely neurons are unique, and there must be a reason why minds associate with brains? This assumption, which permeates discussions of AI and philosophy of mind, is becoming increasingly difficult to hold for several reasons. First, evolutionary developmental biology shows how brains evolved gradually from other cell types. Indeed, the molecular mechanisms that set up the bioelectric networks of the brain are ancient—existing in all cells of the body and having their origin in bacterial biofilms [8–11]. Phylogenetically, ancient traits of all kinds are carried forward and even implemented by new substrates (i.e. degeneracy)—so, why not the mechanisms underlying mindedness? Second, memory, creative problem-solving, decision-making and numerous other aspects of active agency have now been observed in cells, tissues, slime moulds and even small molecular networks [12–16]. Advances in biophysics, cell biology and behavioural science are revealing deep evolutionary commonalities between the mechanisms and algorithms of neuroscience and other fields such as developmental biology, for example, in the study of cell groups as a collective intelligence that navigates anatomical space [17,18].

Thus, the emerging field of diverse intelligence reveals that the focus on brains, and on the readily observable movement of organism-sized (cm, m) objects at behavioural time scales (ms, s) through the three-dimensional world, is a result of our evolutionary firmware. Efforts to expand from this neuro- and anthropocentric myopia are underway via research on the problem-solving competencies of a wide range of living and synthetic agents, which is helping to erode our limited evolutionary firmware for visualizing minds in unconventional embodiments [19]. To date, however, all of the work in this field has focused on externally observable behaviour—problem-solving, or the 'easy' problem of consciousness [20]. What of the hard problem—how to explain the inner perspective and elucidate its relationship to the brain? That current theories of consciousness (ToCs; [21,22]) focus on correlates, markers, metrics, physical interactions, objective referents and other events with which to infer conscious states rather than pointing directly at subjectivity reflects a long-standing habit in the field of defining consciousness in terms that presuppose what it sets out to explain. But if that is the standard when evaluating consciousness in humans, the same standard should be applied to all systems; neglecting to do so *a priori* is tantamount to begging the question.

Here, we explore the state of consciousness research with a focus on ToCs and inquire about the implications of recent breakthroughs in biology for theories that make a claim about what is necessary and/or sufficient for consciousness. Specifically, we analyse popular current ToCs to ask: What features of the theory specifically pick out brains as a privileged substrate of inner perspective, or do the features emphasized by each theory occur elsewhere, in natural bodies and perhaps even in the world of engineering? We find that most ToCs rely on functional

principles that are not specific to neuronal networks, which means that their claims about consciousness should apply to a wide range of unconventional substrates. We find few specifics about why neuronal assemblies, in particular, would be an exclusive substrate for consciousness and suggest that the focus on brains is more driven by convention and understandable limitations of imagination than by any specific content of existing ToCs. Thus, it is interesting to explore the relationships of various ToCs with the nervous system *per se* and the consequences of taking seriously the features they emphasize when they occur outside of brains.

1.1. Theories of consciousness as tools to counteract pervasive mind-blindness

Specific ToCs offer long-awaited explanatory and predictive power amid a history of confusion about the nature of experience. By what standard can a ToC be assessed as useful? Insofar as it can reliably point to subjects in the world, a ToC accomplishes its most basic function. But what if the theory can only reliably point to a particular type of subject? The validity of any theory that purports to explain how and where experience manifests is impacted not only by its positive identifications of mind but also by the number of true minds that it leaves unidentified. Though practical distinctions have always been made to classify the contents of experience, there is yet no consensus definition of consciousness or ToC that accounts for its mechanisms. We usually have no problem being epistemically confident about our own inner perspective. However, the existence of *other* minds with private thoughts and feelings may be fundamentally unknowable [23–26]. Still, humans share a robust intuition that we live among other minds in a shared environment [27]. That others speak, gesture, respond and emote in ways that mirror our own behavioural correlates of consciousness reinforces a theory of mind that underlies the social and moral systems humans have created and refined over hundreds of thousands of years [27], and the ancient biological firmware that tuned our agency detectors for specific kinds of behaviours in visible space. Whether it reflects reality or not, humans engaged in world modelling infer the existence of other embodied minds within their environments, and this leap of faith (and its attendant limitations) is a normal feature of human psychology.

Mind-blindness—or a failure to form a theory of mind in the presence of other humans—is a defining characteristic of some common neuropathologies but is relatively rare in the general population [28]. Indeed, most humans routinely infer the presence of other minds to inform their social behaviour and ethics. However, we suggest that all humans are insensitive to a wide range of unconventional minds, without sensing any kind of deficit, in the same way that we are insensitive to huge swaths of the electromagnetic spectrum without feeling impaired or appearing as such to other humans. If our default cognitive apparatus was significantly mind-blind to major categories of unconventional beings, we would not know about it without developing tools in the same way that physics and a useful theory of electromagnetic waves opened our awareness to the existence of X-rays, radio waves and other signals that our default configuration cannot perceive. This analogy also works because light and X-rays simply *do not seem* like what happens when you wave a magnet up and down—on a superficial level, they appear quite different indeed. And yet, a rigorous formalism allowed us to overcome our limited intuitions and see electromagnetics in all of its guises. Is there a reason to hypothesize that the same situation awaits us with respect to the space of possible minds?

Our apparent inability to model worlds full of unconventional minds could indicate that such a capacity was orthogonal to fitness outcomes throughout our evolutionary history. However, it is more likely that mind-blindness confers fitness advantages in the same way that other distortions of perception are thought to have increased survival and reproduction relative to unfiltered windows to reality [29,30]. Consider the example of predation. While searching for and capturing prey may depend on an ability to predict the actions of prey (which could benefit from a theory of mind), killing and eating prey (in many cases, while the organism is alive and struggling for survival) are unlikely to be advantaged by an awareness of other minds

and their analogous subjective states. Being aware of another animal's pain, fear, suffering or distress of any kind in a way that registers as a reflection of one's own emotional state is unlikely to benefit predators, whose survival is dependent on regularly killing and eating other organisms. Therefore, some minds may be perceptually siloed from others by virtue of their competitive advantage. On the other hand, it is conceivable that some minds were not shaped by similar selective pressures and readily engage in world modelling that includes or even privileges awareness of other embodied minds in the environment. In any case, we propose that regardless of the origins and pressures that shaped our limitations with respect to modelling the full spectrum of agency in our world (and others' worlds), we now have the opportunity (and responsibility) to go beyond those native limitations as we have surpassed so many others.

Developmental biology and evolutionary theory emphasize the continuity of conserved mechanisms and algorithms spanning slowly and gradually from single cells (such as microbes and fertilized eggs) to adult metacognitive humans and the gradual modification of generic cells and cell networks into neurons (figure 1). The gradual self-construction of the body suggests the natural possibility of the gradual appearance of consciousness (and thus its presence, in degrees, in substrates different from an adult brainy animal), as perhaps reflected in Turing's interest in synthetic intelligence and the spontaneous patterning of embryonic morphogens [1,41]. In addition, recent advances in bioengineering, synthetic biological intelligence, AI, brain-computer interfaces and robotics have fundamentally challenged our mind-inferencing strategies. The interoperability of life with engineered components is revealing a wide option space of hybrid beings that do not fit into classic life/machine categories ([42]; figure 2). Thus, whether or not the behaviour patterns of 'artificial' and hybrid systems reveal the presence of an experiencing subject has become a topic of great interest to many. ToCs, as principled theories for where one can expect inner perspective, promise to enable technologies to reveal heretofore unrecognized vistas—like tele-spectroscopy, which revealed familiar earthly elements in then-unexpected locations: celestial objects. To overcome the limitations of a human-centred theory of mind and increase true positive identifications of unconventional manifestations of consciousness in the environment, ToCs must be sufficiently universalizable to account for other minds. Are current ToCs formulated with other minds, well, in mind?

1.2. Why neural correlates fall short as consciousness criteria

Consciousness cannot be directly measured; therefore, correlates have been used as indirect signs of an underlying mind in the same way that the presence of a black hole is inferred on the basis of how light is bent or 'lensed' by distortions of space-time. Many contemporary ToCs refer to 'neural correlates' of consciousness (NCCs), which are a set of conserved relationships between human brain structure-function and self-reported experiences [48]. Why 'neural correlates' as opposed to, simply, 'correlates'? Following localized brain lesions, neural stimulation or the use of psychoactive compounds, humans report highly conserved experiential phenomena that map onto common brain regions as revealed by neuroimaging [48]. While its flow may be interrupted by sleep or altered by drugs, anaesthetics, rituals and environmental forces, the evidence indicates that consciousness is a shared feature of human brains with stereotyped structure-function correlates. Because brain morphology is highly conserved (i.e. most brains display the same gyri, sulci, nuclei, tracts, circuits, networks, etc.), NCCs can be generalized across members of the human species, allowing us to overcome weaker forms of mind-blindness in clinical settings. Until the development of brain imaging, immobile and seemingly unresponsive coma patients were lumped into the category of non-conscious systems with most organisms, robots and the deceased; however, with NCCs, new degrees of distinction could be made, including the confirmation of a minimally conscious state [49–52]. NCCs predict the existence of recondite minds on the basis of neural activations alone if the same activations were associated with the reported experiences of others. This mind-detecting strategy, termed

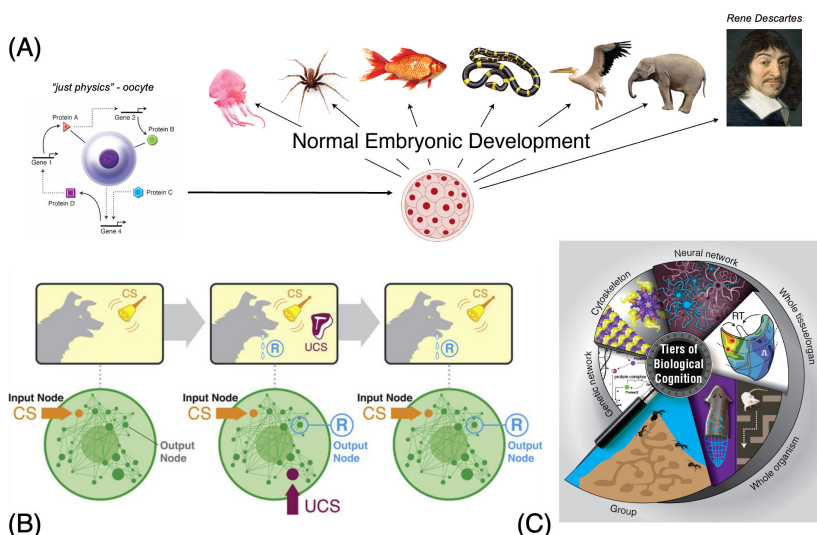


Figure 1. Self-assembly of conscious beings from the agential material of life. (A) Each of us is the product of a transformational process (embryogenesis) that leads from a quiescent oocyte well encompassed by the laws of chemistry to a complex meta-cognitive agent amenable to behavioural science or even psychoanalysis. This journey across the ‘Cartesian cut’ is slow and gradual, with no place for a bright line during which physics suddenly becomes mind. Generic cells become neurons, some electrophysiological networks speed up to become nervous systems and motility in three-dimensional space appears, whereas before there was only navigation of metabolic, gene expression and anatomical spaces. (B) Aspects of cognitive function appear long before neurons and are universal, not even requiring cell networks or whole cells. For example, modelling and wet lab studies are beginning to suggest that unicellular organisms [31,32], chemical networks within all cells and a wide range of materials may be capable of several different kinds of learning [32–36] and even probabilistic inference [37,38]. (C) Biological material in general has problem-solving competencies and agents at every level of organization, from the active matter within cells to tissues and organs navigating physiological spaces and learning from experience [13,39]. Panel A taken with permission from [40]. Panel B taken with permission from [34]. Panel C taken with permission from [17]. All images courtesy of Jeremy Guay of Peregrine Creative.

‘reverse inference’ [53], is a practical solution to mind-blindness when the potential subject in question is a human with a brain.

Unfortunately, any movement across the phylogenetic tree decreases the predictive power of reverse inference because NCCs are fundamentally *correlates* and can only be used to infer consciousness in precisely the same way that behaviours motivate the same inferences. If the associations in question are non-generalizable, such as involving very specific nervous systems with defined organizations, the logic of reverse inference breaks down. While many ToCs have highlighted the relevance of NCCs, their poor generalization to non-human subjects limits their utility. How would an NCC-dependent ToC map onto the distributed nervous system of a jellyfish or the ganglia of a mollusc? What can current ToCs offer in terms of predicting a capacity for experience when comparing three-layered and six-layered cortices or their intermediates? Unless a ToC can account for differences in neural organization, it will return the same predictive errors that human intuitions commit when assessing the mental status of anything other than a human. Some authors have suggested the use of perturbational complexity as a measure of consciousness, which, despite its focus on the responses of neural tissues, has the potential to transcend a historical reliance on specific NCCs because, in principle, other systems can display brain-independent perturbational markers that predict conscious experience [54,55].

Despite the long-standing absence of any direct measure of consciousness and strong philosophical reasons to doubt its existence in others, humans often extend a theory of mind to infer non-human animal sentience [56]. Interestingly, the degree to which a non-human animal shares superficial, human-like characteristics is predictive of our willingness to infer

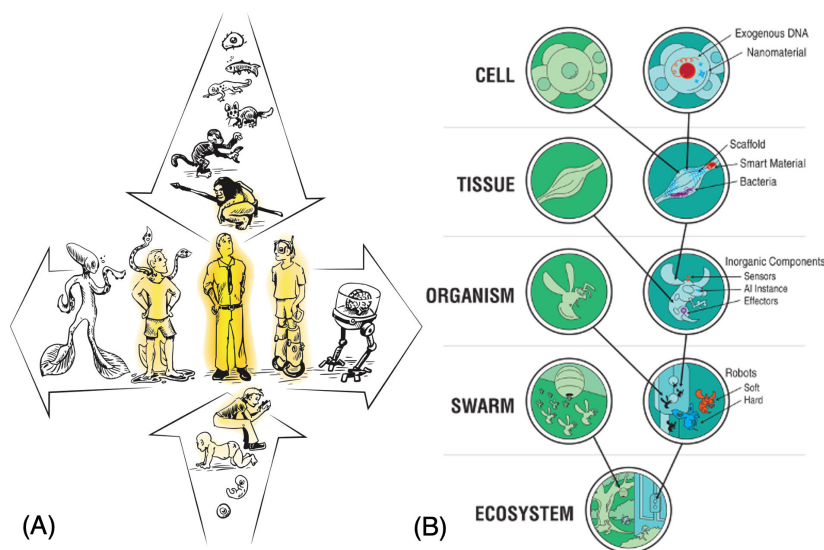


Figure 2. The continuum of the natural and the engineered. (A) The ‘human’ that features so prominently in conventional philosophy of mind is at the centre of two continua. First (vertical) is the set of gradual changes on the evolutionary and developmental time scale that connects us to single cells and subcellular matter before that. The sense of consciousness we attribute to humans (represented by the agential glow) must, given the facts of developmental and evolutionary biology, be explained with respect to its degree and kind in closely connected forms. The continuity and transformation of mental properties from very minimal origins is the null hypothesis, and theories proposing the appearance of binary categories (‘is it conscious or isn’t it’) would need to argue convincingly and explicitly for mechanisms of emergent phase transitions. Second (horizontal) is a spectrum of continuous change that can be (and is being) implemented, both biologically and technologically, resulting in a wide array of composite, chimeric beings that are not easily categorized within or outside of ‘human’ minds. This means that our theories of consciousness should not be human-centred but rather encompass a wide and continuous space of minds [43–47]. (B) Another key feature of biological organization is that it can be readily hybridized with engineered, designed components at every level. This interoperability of life with novel structures means that discussions of consciousness cannot focus on ‘life versus machine’ but must accommodate a rich space of chimeric forms with features not explainable by evolutionary history or selection pressures. Panel B taken with permission from [17]. All images courtesy of Jeremy Guay of Peregrine Creative.

subjectivity on the basis of their behaviours [57,58]. Once again, reverse inference is used to bridge the epistemic gap (i.e. analogous behaviours and observable traits as predictors of mind). Prior to a scientific study of animal consciousness, moral considerations were granted to many non-human animal species, and most people are quite happy to assume that their pets lead rich, experiential lives with felt states of pleasure and pain. In many countries, non-human animal species, such as dolphins and chimpanzees, enjoy personhood status [59,60] that sets them apart from other living systems, including plants, insects and microorganisms—which together constitute over 99% of Earth’s biomass [61]. In other words, less than 1% of living systems on the planet are granted special status and consideration over the vast majority due to their presumed subjectivity, even though the rest have long been known to exhibit different degrees of learning, decision-making, predictive capacity, etc. [13,62–65]. Notably, much less than 1% of living systems on Earth possess a nervous system. If even a minority subset of aneural life is conscious—perhaps a single plant or animal species—the NCC standard will fail to account for their experiences, generating false negative errors wherever it is applied.

While recent developments in synthetic biological intelligence and AI research have accelerated discussion around the topic of unconventional minds, the emergence of neural-robotic hybrids in the late twentieth century represented a major branching point in the conversation. That cultured neural networks could be coupled with robotic or virtual ‘bodies’ to solve real-world problems challenged long-standing dichotomies between the living and

non-living or the agential and automatic [66–68]. It was quickly realized that disembodied brains were dysfunctional but could be normalized with stochastic inputs or response-contingent feedback [69–71]. Now, there is evidence to suggest that cultured neural networks in closed-loop feedback systems can optimize their outputs towards many types of defined goal states without external programming [72]. Importantly, these displays of self-organized problem-solving are achieved without specialized, genetically encoded brain circuits. That is, embodied cultured networks demonstrated that, fundamentally, there is no intrinsic, universal value to any particular tract system, nucleus or circuit outside of its default context. Rather, a sufficiently plastic neural network can be shaped to accomplish any number of tasks. Once considered exclusive to human brains, cognitive functions such as learning, attention and decision-making have, over time, been incrementally extended to living and non-living systems alike. While it is possible that any one or perhaps all of these features of cognition could be displayed without an experiential correlate, the stakes are high—not knowing whether a physical object with which you are engaged is conscious is an ethical risk. This is especially relevant now that a much wider variety of beings—biobots [73–76], cyborgs [77–83] and minimal active matter systems [84–88]—are coming online and beginning to be bioengineered. ToCs will need to be applied to an ever-increasing number of organizationally unique systems with potential minds [17,42,89].

ToCs are important. Whether or not they hold construct validity (i.e. how well the ToC not just predicts the presence of a subject but actually describes true features or mechanisms of consciousness), they enable us to make meaningful distinctions that inform decisions related to lifestyle, ethics, public policy and law. Frameworks must now be developed to identify unfamiliar minds in unconventional spaces that betray long-standing neurocentric assumptions [17,65,90]. Here, we survey contemporary ToCs and explore their suitability to identify other kinds of minds in diverse environments. How many of them are actually neuron-specific? While many ToCs have already adopted a panpsychist, functionalist or universalized approach, some still focus exclusively on the brain, even though nothing about their ‘secret sauce’ is actually unique to neurons.

2. Neural tissues as non-exclusive specialists of cognitive function

Because ToCs that reference NCCs or any specific neurobiological details will likely fail to generate a true positive prediction of unconventional minds, the role of the brain within the cognitive landscape must be revisited. Neurons—and specifically, pyramidal cells—were once described by Ramón y Cajal as ‘the butterflies of the soul’. The implication, of course, was that neural cells conferred the properties of minds, which remains a central assumption of modern neuroscience. However, there is no single feature of the neuron that is not also displayed by some other cell, living system, inorganic material or natural process (figure 3). Whether it is a capacity for long-range signalling, cell shape plasticity, chemical communication, electrotonic coupling, electromagnetic sensing, membrane polarization, chemotaxis, galvanotaxis, ephaptic coupling or networking, the neuron holds no exclusive capacity or function [93–95]. While the label ‘neuron’ is most commonly applied to cells with the ability to form very long point-to-point connections, many other cell types possess degrees of this architecture via tunnelling nanotubes and other structures (reviewed in [96]), including pigment cells that can make long projections, reaching a significant proportion of the way across the entire body in tadpoles [97].

Many living systems display electrochemical patterns that are homologous to those of neurons, including the electromagnetic fields that have been hypothesized to be important for consciousness [98–104]. Most bodily cells regulate membrane-bound ion channels to maintain a polarized electrical state; however, neurons display notably hyperpolarized resting potentials (−70 mV). Cardiomyocytes, which are similarly hyperpolarized at rest (−90 mV), generate spontaneous depolarizations or ‘action potentials’ [105] with refractory periods and other dynamics that are similar but not identical to those of neurons (and participate in a

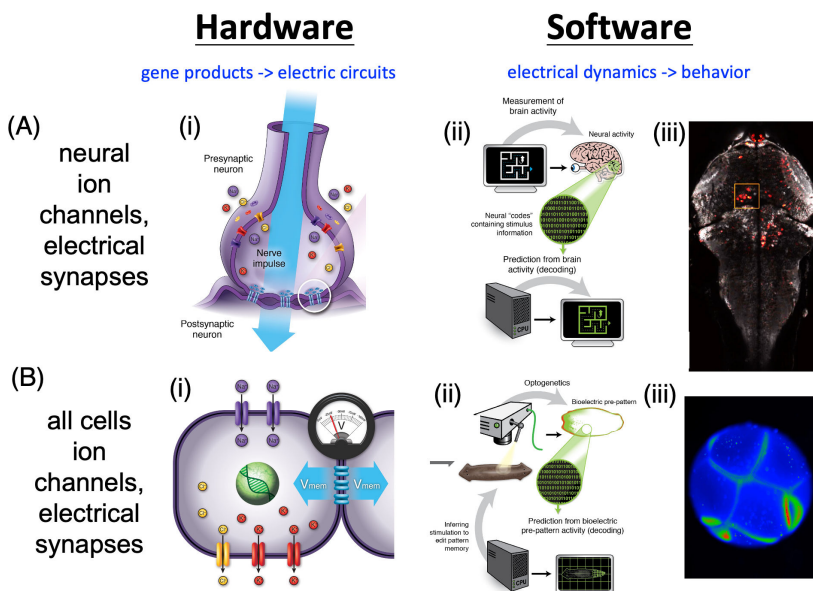


Figure 3. Strong symmetry between neural and non-neural cognitive substrate. (A) In the nervous system, the hardware consists of a network of cells connected by chemical and electrical synapses (i); the goal of neural decoding (ii) is to read the physiological states of the brain (iii) bioelectric imaging of a living vertebrate brain) and decode them to infer the cognitive content of the mind embodied by this network—to be able to extract the memories, goals, preferences and conscious state. (B) The powerful properties of bioelectric networks that enable memory, re-entrant network architectures and integration of information across space and time are not unique to neurons: all cells in the body have ion channels for generating spatio-temporal patterns of resting potential and gap junctions (electrical synapses) for shaping those patterns in light of ongoing processing (i); precisely the same tools, ranging from optogenetics to computational pipelines to crack the somatic bioelectric code (ii), are being used to decode the goal states, memories and navigational behaviours of bioelectric patterns (iii) voltage sensing dye showing dynamic gradients in the early frog embryo) during decision-making and problem-solving in the anatomical morphospace. Panels A(i), B(i) and B(iii) taken with permission from [40]. Panels A(ii), B(ii) taken with permission from [91]. Panel A(iii) taken with permission from [92]. Image in panel B(iii) is by Dany Adams, Levin lab. All other images are courtesy of Jeremy Guay of Peregrine Creative.

phenomenon known as cardiac memory [106,107]). Immune cells can be sensitized, display learnt responses and encode long-term memories reflective of a history of exposure to specific pathogens [108], just like cells of the nervous system form long-term memories reflective of a history of sensory inputs. Neurotransmitters, such as glutamate and serotonin, are expressed throughout the body, outside the brain [109] and by many plant species as chemical signals for cellular communication [110]. Incidentally, plants display anticipatory responses and even game-theoretic decision-making capabilities that weigh the availability of resources with the kin status of neighbouring organisms, determining the expression of competitive or cooperative actions [111]. Unsurprisingly, some authors have called for a more inclusive definition of ‘nervous system’ that captures a spectrum of signal generation, transmission and processing in multicellular systems [112]. Paramecia [32,36,113] and slime moulds [114–116] can be conditioned and even solve problems without nervous tissues. We previously demonstrated that non-associative learning, such as habituation, is a commonplace function that is not dependent on any specific biological substrate or cell type [117,118]. Several authors have since suggested that learning is a universal property shared by cells, generally [14,15,119–123].

Dynamical properties of neurons are also displayed by simple molecular interactions at sub-cellular scales [124]. Indeed, phospholipid membranes alone display memcapacitive properties that recapitulate long-term potentiation without cells or synapses [125]. Water at an interface generates negative potential differences (voltage) within the range of resting membrane potentials (−100 mV) that can be modified by the addition of physiological ions

and detected hundreds of microns away from the surface [126]. Microtubules and their special properties have been suggested to be critical for consciousness [127–130], but cytoskeletal structures, such as microtubules, are present in all cells and play an important role in cellular and multicellular decision-making and collective behaviour [131–134]. Moreover, microtubules in a dish spontaneously align with and migrate along electric fields [135,136] and polymerize as branched networks with dendrite-like arbours. Incidentally, branching patterns of neurites, plants and blood vessels conform to the same scaling laws [137–139]. Even inorganic substrates, such as iron bars, are subject to conditioned hysteresis responses [140], and silver nanowires display self-assembling properties in the presence of electric current that optimize patterning with properties of learning and memory [141]. Interestingly, the current densities, firing frequencies, backpropagation responses and saltatory-conductive properties of neurons are also displayed by lightning strikes between the Earth's surface and its atmosphere [142].

Neither the composition nor the dynamics of neurons is sufficiently unique to justify their special status as mind generators. Either individual cell properties are insufficient or many types of simple systems are similarly capable of generating conscious states. The third possibility is that brains deserve exclusive status because of their connective properties, local circuitry and network architectures. It is often suggested that structural complexity is what sets brains apart from other systems. However, contrary to popular assumption, a network architecture with sparse rather than dense connectivity is likely to specialize brains as implementers of cognition [143]. Notably, feedforward networks are unlikely to generate consciousness [52]. Feedback mechanisms, including reverberation by re-entrant circuits, are thought to be requirements for consciousness [144,145]. From a functionalist perspective, the organizational principles are more relevant than the substrates, and neural tissues represent one of many possible ways to implement cognitive functions by organizing matter into structured networks that transduce, process and transmit patterned energy as information. With many natural and synthetic examples of re-entry, feedback, network sparsity and plasticity outside the nervous system, there is good reason to doubt that brains occupy a privileged position over other systems with analogous organizational features. Here, we examine several contemporary frameworks from a functionalist perspective, with the aim of universalizing ToCs to interact with unconventional substrates and scales.

3. Theories of consciousness to map diverse embodied minds

A universalizable ToC should be substrate-independent, scale-invariant and organization-invariant [20], with the implication that *mind* is multiply realizable [89,146]. Unfortunately, when the properties of cognitive systems are divorced from their biological substrates, it can be difficult to know where to apply inferencing strategies and deeply unintuitive questions arise: what does it mean to be conscious in spaces other than the three-dimensional space at human spatio-temporal scales? Beyond the neural variety, what kinds of correlative phenomena can reliably serve as indirect measures of consciousness? What kinds of perturbations or physical interactions can reliably induce changes within a system that indicate the likely presence of a mind? Without the grounding of a testable ToC, little progress can be made. As an initial step towards mapping minds in unconventional spaces, we reviewed prominent theories of consciousness as recently selected by Seth & Bayne [21] and substituted neurocentric language with aneurocentric or generic terms. Expanded, aneurocentric versions of the primary claims of each theory were then constructed with language reflecting inclusion of different substrates, scales and organizations (see table 1). Common terms such as 'sensation' or 'sensory' were replaced with generic afferents such as 'input'. Likewise, terms such as 'motor' were replaced with generic efferents, such as 'output'. Similarly, terms such as 'brain' or 'neuron' were replaced with generic terms for potential loci of consciousness, including 'system' or 'processor', which we consider potentially interchangeable with 'cell', 'core', 'hub' and/or 'node', depending on the context of the specific ToC.

Table 1. Aneurocentric formulations of prominent theories of consciousness. The table contains the main ToCs (taken after [21]); *italics* indicate the words that were generalized to show how the theory applies beyond brains. In many cases, no changes needed to be made. This illustrates how many ToCs are actually not about brains specifically but call out aspects that are relevant to diverse cells, systems and collectives.

theory	primary claim	references
active inference	original formulations (verbatim from [21])	consciousness depends on temporally and counterfactually deep inference about self-generated actions [147,148]
	substitutions	none
	expanded aneurocentric formulation	consciousness arises from an embodied system's continuous predictions about inputs, adjusting internal states and outputs based on prediction errors to minimize differences between expected and actual inputs
attention schema theory	original formulation	consciousness depends on a <i>neurally</i> encoded model of the control of attention [149]
	substitutions	<i>neurally</i> → information-processing system
	expanded aneurocentric formulation	consciousness arises from an information-processing system assigning high degrees of certainty about the claim that the system itself contains an attention schema or subject
attended intermediate representation theory	original formulation	consciousness depends on the attentional amplification of intermediate-level representations [150,151]
	substitutions	none
	expanded aneurocentric formulation	consciousness arises when intermediate-level or partially processed information becomes amplified by attentional resources and can enter working memory
beast machine theory	original formulation	consciousness is grounded in allostatic control-oriented predictive inference [152–154]
	substitutions	none
	expanded aneurocentric formulation	consciousness is the effect of combining a subjective frame, or internal reference point generated by constantly updated representations of internal states, with perceptual content evoked by external stimuli
dendritic integration theory	original formulation	consciousness depends on integration of top-down and bottom-up signalling at a <i>cellular</i> level [155]
	substitutions	<i>cellular</i> → processor
	expanded aneurocentric formulation	consciousness depends on a coupling between bottom-up and top-down processing that generates a reverberating or looped function
dynamic core theory	original formulation	consciousness depends on a functional cluster of <i>neural</i> activity combining high levels of dynamical integration and differentiation [156]

(Continued.)

Table 1. (Continued.)

theory	primary claim	references
	substitutions	<i>neural</i> → processor
	expanded aneurocentric formulation	consciousness arises from an integrated cluster of high-complexity re-entrant processors, or a compositionally dynamic 'core', within an information-processing system that is much more strongly interactive with itself than with other parts of the system
electromagnetic field theory	original formulation	consciousness is identical to physically integrated, and causally active, information encoded in the <i>brain's</i> global electromagnetic field [99,157]
	substitutions	<i>brain</i> → information-processor
	expanded aneurocentric formulation	consciousness depends on information-processing activities by processors interacting with the complex interference patterns of time-varying electromagnetic field oscillations
global workspace theories (GWTs)	original formulation	consciousness depends on ignition and broadcast within a neuronal global workspace where frontoparietal cortical regions play a central, hub-like role [158–160]
	substitutions	<i>neuronal</i> → information-processing system <i>frontal/parietal cortical regions</i> → processors
	expanded aneurocentric formulation	consciousness arises from the attention-gated integration of information within an exclusive global workspace or 'hub' that uses working memory and can selectively broadcast to multiple processors simultaneously that are within the system but outside of the hub
higher order theory (HOT)	original formulation	consciousness depends on meta-representations of lower order <i>mental states</i> [161,162]
	substitutions	<i>mental states</i> → states
	expanded aneurocentric formulation	consciousness is the higher-order, meta-representational monitoring of the processing activities associated with lower-order states
information closure theory	original formulation	consciousness depends on non-trivial information closure with respect to an environment at particular coarse-grained scales [163]
	substitutions	none
	expanded aneurocentric formulation	consciousness is a process Y that forms information closure with a stochastic process X of which it is a coarse-grained product
integrated information theory (IIT)	original formulation	consciousness is identical to the cause–effect structure of a physical substrate that specifies a maximum of irreducible integrated information [164–166]

(Continued.)

Table 1. (Continued.)

theory	primary claim	references
local recurrency	substitutions	none
	expanded aneurocentric formulation	consciousness corresponds to the level of integrated information within a system, quantified by a measure (ϕ) where higher integration indicates richer, more coherent experiences
	original formulation	consciousness depends on local recurrent or re-entrant cortical processing and promotes learning [167,168]
multiple drafts model	substitutions	cortical $\rightarrow$ processor
	expanded aneurocentric formulation	consciousness arises from recurrent or re-entrant feedback at points of interaction between higher and lower level processors
	original formulation	consciousness depends on multiple (potentially inconsistent) representations rather than a single, unified representation that is available to a central system [169]
neural Darwinism	substitutions	none
	expanded aneurocentric formulation	consciousness consists of highly parallelized, multi-track processes of informational interpretation and elaboration with continuous editorial revision that are not represented as unified narratives
	original formulation	consciousness depends on re-entrant interactions reflecting a history of value-dependent learning events shaped by selectionist principles [170,171]
neural subjective frame	substitutions	none
	expanded aneurocentric formulation	consciousness is a re-entrant interaction between perceptual processors and relay nodes along their pathways, conferring selective advantages by linking current inputs with a history of weighted input values
	original formulation	consciousness depends on neural maps of the bodily state, providing a first-person perspective [172]
neuro-representation alism	substitutions	neural $\rightarrow$ system/processor bodily $\rightarrow$ environmental interface
	expanded aneurocentric formulation	consciousness is a dynamic frame of reference that is dependent on continuously updated maps of the system's internal states that do not directly interface with the environment, but which are impinged upon by processors of external stimulation
	original formulation	consciousness depends on multi-level neurally encoded predictive representations [173]
	substitutions	neurally $\rightarrow$ non-environment/system

(Continued.)

Table 1. (Continued.)

theory	primary claim	references
	expanded aneurocentric formulation	consciousness is a multi-modal, context-dependent survey that predicts representations of input stimulation (i.e. superinferences) from the environment and enables goal-directed actions upon the environment
orchestrated objective reduction	original formulation	consciousness depends on quantum computations within <i>microtubules</i> inside <i>neurons</i> [174]
	substitutions	<i>microtubules</i> → qubit reservoirs/processors <i>neurons</i> → cell/system
	expanded aneurocentric formulation	consciousness is the terminal product of complex, quantum computations performed by oscillating electromagnetic dipoles as qubits that undergo state reduction with a definite outcome
predictive processing	original formulation	perception depends on predictive inference of the causes of <i>sensory</i> signals; provides a framework for systematically mapping <i>neural</i> mechanisms to aspects of consciousness [175–178]
	substitutions	<i>sensory</i> → inputs <i>neural</i> → internal/processor
	expanded aneurocentric formulation	consciousness arises from the process of actively testing hypotheses as internal predictive models of expected inputs against streams of actual inputs
sensorimotor theory	original formulation	consciousness depends on mastery of the laws governing <i>sensorimotor</i> contingencies [179]
	substitutions	<i>sensorimotor</i> → input-output/embodiment
	expanded aneurocentric formulation	consciousness is a system's capacity to display embodied engagement (output) with stimuli (input) from the environment without the necessity of internal representations
self comes to mind theory	original formulation	consciousness depends on interactions between homeostatic routines and multi-level interoceptive maps, with affect and feeling at the core [180,181]
	substitutions	none
	expanded aneurocentric formulation	consciousness is a state that occurs when a system contains an informationally closed internal representation of itself, situated relative to an external environment.
self-organizing meta-representational theory	original formulation	consciousness is the <i>brain's</i> (meta-representational) theory about itself [182]
	substitutions	<i>brain</i> → system
	expanded aneurocentric formulation	consciousness occurs in systems that can attend to their first-order internal states, which are generated by impinging external stimuli, and have learnt to value some states over others

(Continued.)

Table 1. (Continued.)

theory	primary claim	references
unlimited associative learning	original formulation	consciousness depends on a form of learning that enables an <i>organism</i> to link motivational value with stimuli or actions that are novel, compound and non-reflex inducing
	substitutions	<i>organism</i> → system
	expanded aneurocentric formulation	consciousness arises when a system's states are processed by high-level integrating units, learning about itself in an open-ended way with compounds of paired patterns of stimuli and actions

When the substrate- and scale-dependent contents of each theory are separated from their functional principles, it becomes clear that, among the most prominent ToCs, there are only a handful of distinct concepts or themes including: (i) predictive modelling, (ii) enactivism/ecological interactions, (iii) re-entrancy or looped feedback, (iv) meta-representations, (v) attentional gating/monitoring, (vi) emergence from computation, (vii) integration of information, and (viii) coarse-graining. Some ToCs are mutually incompatible; however, viewed through a functionalist lens, many more display broad conceptual overlap. While there is no doubt that particular brain regions (e.g. cortices, brainstem nuclei), networks (e.g. default mode network, attentional, sensorimotor) and circuits (e.g. corticothalamic), as well as neurophysiological processes (e.g. gamma oscillations) are at least necessary for human brains to generate consciousness, efforts to extend ToCs to other minds must involve a generalization and analysis of the biological details of such theories to test mappings of the theory onto functions of all cells (and thus organs). Aneurocentric formulations of ToCs are thus testable while enabling a comparative analysis of consciousness across very different systems.

4. Implications of consciousness in agents living in unconventional spaces

Observable phenomena by which we typically infer some degree of consciousness, including learning, decision-making, navigation, eye tracking and goal-directedness, typically occur in three-dimensional behavioural space at familiar, human-centred scales. However, it is now known that many cells, tissues and aneural organs can do all of these things in metabolic, physiological, transcriptional (gene expression) and anatomical spaces ([19]; figure 4). They formulate and pursue goals, solve new problems they have not seen before, exhibit taxis and aversive behaviour with respect to regions of their state space and align components towards large-scale goals (reviewed in [14,64,65,119,134,191–193]). Indeed, it appears that the interesting properties of brains arose as a gradual evolutionary pivot of fundamental capacities (including active inference and homeodynamic goal-directedness) across problem spaces [93]. While difficult for us to visualize directly, because of our sense organs and our own evolved theory-of-mind firmware focused on a fixed range of embodiments and behaviours involving obvious motility, it is essential to let advances in science expand our native perspective. To the extent that research in basal cognition and diverse intelligence reveals behavioural competencies in unconventional spaces, we must be open to the applicability of ToCs to these scenarios. As seen in table 1, and consistent with the very high conservation of mechanisms and algorithms between neurons and non-neural cells, no existing ToC rules out aneural substrate. For precisely the same reasons, we routinely entertain the possibility of consciousness in brainy animals based on their behaviour and its biophysical underpinning, we must consider the possibility that cells, tissues, organs, organoids, biobots and a wide range of chimeric cyborg/

hybrots architectures could have inner experience as they intelligently navigate, strive, achieve and suffer in their own worlds of possibility. And indeed, as known from advances in the field of morphological computation [194–196], the structure of an agent's problem space has a massive impact on their embodiment and cognition, which suggests that minds navigating unconventional spaces may not be easy for us to detect unaided.

5. Developing research programmes

This perspective, grounded in developmental and evolutionary biology of the biophysics underlying neural networks, has a number of implications for a research roadmap. The use of tools and concepts of neuroscience and behavioural science is already paying off in terms of empirical discoveries and new experimental vistas in these fields, for example, in the use of the collective intelligence of cells navigating anatomical morphospace to impact birth defects, regeneration and cancer (reviewed in [91,197,198]) and the investigation of learning and memory in gene-regulatory networks with possible impacts on a wide range of pharmacological use cases [33,34,199,200]. But to date, these have all been strictly third-person perspective, standard science. It is time to extend consciousness studies and philosophy of mind to cell biology, regenerative medicine and bioengineering.

An obvious next step is to use metrics from causal information theory and IIT (integrated information theory [201,202]) to study the information flows during behaviour in non-neural agents [203]. This has already begun, in work to measure information architecture [204] and causal emergence metrics in Xenobots [205] and gene-regulatory networks [206]. Many other substrates, such as calcium and bioelectric signalling, during embryonic and regenerative morphogenesis [207], carcinogenic transformation [208,209] and self-assembly of biobots [73,74,210,211], remain to be tested. We also envision the development of classic tools, such as the mirror test and variants of the Turing test, in spaces that make them applicable to cells and similar unconventional agents.

One of the key lynchpins in discussions of both AI and organoid consciousness has been the criterion of embodiment [89,212–216]. Many have argued that engagement with a rich action space via a perception-action loop (including perceptual control [217] or active inference [218–220]) is critical for the formation of true agents with consciousness. We agree, but point out that movement in three-dimensional space is not the only arena for this critical dynamic. It is possible that 'disembodied' organoids, for example, which offer no obvious behaviour in three-dimensional space (because they lack muscles and other effectors), in fact display a kind of 'locked-in syndrome' [221] that makes observers think there is no one inside, whereas they have an active life, solving problems in gene expression, physiological and other spaces. For the same reasons we seek to develop tools to identify inner perspective in human cases of locked-in syndrome or coma [54,55], we must broaden our perspective and create substrate-agnostic conceptual and experimental tools to detect, quantify and characterize these processes across the diverse agential material of life, from sub-cellular molecular networks to organ systems. We hypothesize that the existing tools of the neuroscience of consciousness, combined with virtual reality tools to assist visualization of behaviour in high-dimensional, unconventional spaces by human scientists, will be a powerful combination to begin to overcome our innate inability to see all but a tiny fraction of the endless, most beautiful [222] forms of minds.

By loosening historical constraints on the nature of conscious embodiments and the spaces within which they must navigate, the science of consciousness gains access to a wide range of beings that, while seemingly alien, are functionally familiar. Many questions abound with respect to the kinds of minds that exo-biological life forms would have; while we do not have access to true aliens, we now have the opportunity to try to understand minds that are on the same evolutionary tree as us, and thus perhaps tractable, but will force us to expand and refine our conceptual apparatus because they are not tractable to anthropocentric, brain-focused formalisms. The recognition of possible consciousness in living material more broadly

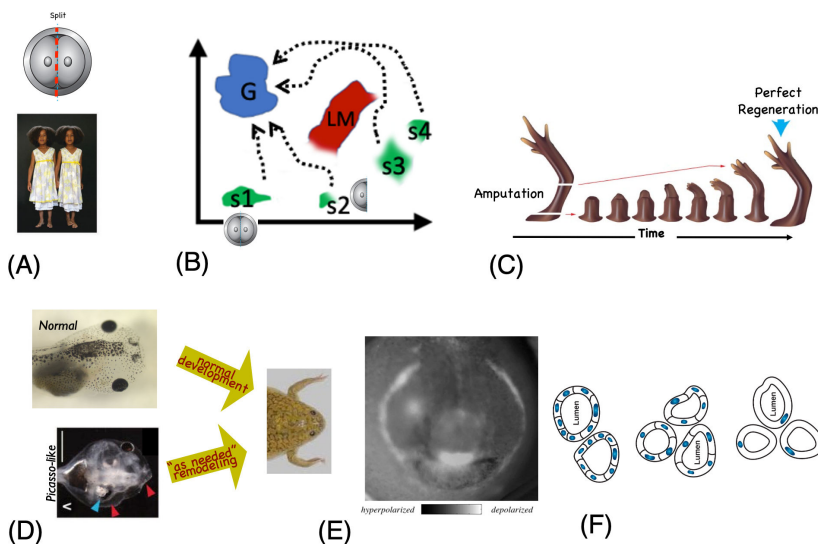


Figure 4. Morphogenesis as behaviour of cellular collective intelligence in morphospace. (A) Human embryos split in half give rise to monozygotic twins (and higher multiples), not half-bodies, illustrating that development is not a hardwired but a flexible, context-sensitive process that can accommodate novel perturbations to reach its goal. (B) More generally, represented in a two-dimensional schematic of morphospace (a high-dimensional space of possible anatomical configurations), normal embryos (S1) can reach the ensemble of goal states (G) representing a normal species-specific target morphology, but in many life forms, so can embryos split in half (S2) or undergo a wide range of other perturbations (S3, S4). This ability to reach their goal from different starting positions in the space, often avoiding getting stuck in local maxima (LM), represents a kind of navigational problem-solving. (C) Some animals maintain this in adulthood, such as limb regeneration in amphibians: the appendage can be cut at any level and then regenerates the precise amount and shape of missing materials, *stopping when a correct limb is produced*. This capacity, along with regulative embryogenesis, is a kind of error minimization loop that is carried out by cells, all of which are aligned towards a large-scale goal in the anatomical space. (D) Another example of flexible pursuit of goals by non-neural cells is the formation of a frog face from the reorganization of tadpole craniofacial organs, which can begin in their normal positions or a scrambled (Picasso-like) state, because the paths they take are not hardwired but adjusted dynamically to reach the correct pattern despite unexpected starting configurations [185]. (E) The goal state, with respect to which cellular effectors operate in these cases, can be stored as a bioelectrical pattern memory, akin to the storage of goal states in neural memory systems [11]; here is shown one example—the ‘electric face’ pattern ([186]; visualized with a bioelectrical reporter dye), which instructs the location of the components of the face. Voltage imaging in non-neural cells now allows the read-out (and increasingly, decoding) of the information patterns in bioelectric networks that guide behaviour towards specific future states. (F) Cross section of kidney tubules in normal versus polyploid newts. As polyploidy increases, it causes an increase in cell size, but the overall structure remains normal because the cells adjust by having fewer cells participate in the process. The right panel shows the most extreme case, where the cells are so big that they use a different molecular mechanism to accomplish the goal (by having one cell wrap around itself). Panel A used with permission from Oudeschool via Wikimedia Commons. Panel C used with permission from [187]. Panel D: top left image used with permission from [188]; bottom left image used with permission from [186]. Right image courtesy of Erin Switzer, Levin lab, and modified after [40]. Panel E used with permission from [186]. Panel F adapted from [189], used with permission from [190]. Panels C and F by Jeremy Guay of Peregrine Creative.

will recalibrate current debates about octopuses, crustaceans, plants, etc., and raise fascinating questions about ethical relationships with this much broader class of beings.

6. Conclusion

Taking the slow, gradual scale-up seriously from single cells revealed by developmental and evolutionary biology makes the continuity thesis the null hypothesis. The high conservation of

mechanisms and behaviours in brains all the way back to pre-cellular material implies that a kind of panpsychism, committed to understanding the scaling and transformation of embodied minds from physical dynamics, is not only viable but should also be the baseline assumption. Indeed, our approach places consciousness on a functional continuum not unlike any other capacity. The claim that an organism (or any system) is or is not conscious is, in our view, equivalent to the claim that an organism can or cannot fly, swim, see, hear or carry out any particular function. Rather than forcing binary categorization, it may be useful instead to ask about the degree to which an organism is able to resist gravity, displace fluids or transduce waveforms. Competing ideas, relying on sharp phase transitions and brain-specific theories, need to specify principled reasons for discontinuities and explain the ‘emergence’ of novel natural kinds.

The state of the art in physiology and diverse intelligence research, combined with the compatibility of current ToCs with aneural substrates, suggests that consciousness may be common throughout the body. Our mind supervenes on a collection of cells, working together by means of a bioelectric network that aligns them towards larger cognitive light cones in abstract problem spaces. That architecture is ubiquitous throughout our bodies and throughout evolution. Thus, consciousness in a collective intelligence made of cells is not a wild claim—indeed, it is the only kind of consciousness we have ever seen, because *each of us is a collective intelligence* (of neurons). For all the reasons discussed above, we can drop the part in parentheses from the list of requirements and get on with the task of understanding collective intelligence in all of its general guises and the ways in which it enables intelligence to come into the world.

But it is often objected: ‘we don’t feel our liver being conscious!’. While that is true, we do not feel each other being conscious either. Indeed, if the liver or its parts were capable of subjective experience, that point of view would be quite independent from the consciousness of a brain with which it shared a body. We suggest dropping the unfounded requirement that a body has only one consciousness, as well as the privileged perspective of the one body organ that can eloquently proclaim its lonely unity by way of the left hemisphere’s capacity for language. Many brain structures are mirrored across hemispheres and can function independently following interhemispheric disconnections, including callosotomy. While perception can clearly be divided into split-brain patients, the evidence for divided consciousness is mixed [223], in part because of a shared body (which makes discernment of dual nature by outside observers, or even by one of the hemispheres, challenging). However, if each hemisphere were reciprocally connected to its own independent body, evidence from hybrid robotics research would predict unique behavioural correlates of consciousness suggestive of a capacity for multiple subjects within a shared body. At this point, there is no evidence that the consciousness of non-brain body organs would be the complex, self-reflexive consciousness enjoyed by the human mind or would be capable of language. But this still leaves a huge swath of the consciousness spectrum, which biology likely occupied in its journey from physics. By expanding into the field of diverse intelligence and unconventional embodiments, we can begin to truly deal with the problem of other minds and the richness of conscious kin both within and around us.

Data accessibility. This article has no additional data.

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors’ contributions. N.R.: conceptualization, writing—original draft, writing—review and editing; M.L.: conceptualization, writing—original draft, writing—review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. M.L. is an advisor to a company called Softmax, which seeks to operate in the space of artificial intelligence.

Funding. We gratefully acknowledge the support of Eugene Jhong and of the Army Research Office under grant number W911NF-23-1-0100. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied,

of the Army Research Office or the US Government. We also acknowledge support from the Natural Sciences and Engineering Research Council of Canada (NSERC: RGPIN-2022-04162) and the New Frontiers in Research Fund Exploration Grant (NFRF-EG: NFRFE-2023-00568).

Acknowledgements. We thank Daniel C. Dennett, Chris Fields, Karl Friston, Pamela Lyon, Anil Seth, Mark Solms and many other members of the community for numerous helpful discussions. We thank Julia Poirier for invaluable assistance with manuscript preparation.

References

1. Turing AM. 1950 Computing machinery and intelligence. *Mind* **59**, 433–460. (doi:10.1093/mind/LIX.236.433)
2. Rosenblatt F. 1967 Recent work on theoretical models of biological memory. In *Computer and information sciences* (ed. J Tou). Washington, DC, USA: Spartan Books.
3. McCulloch WS. 1988 *Embodiments of mind*, pp. 28–402, vol. xxviii. Cambridge, MA: MIT Press.
4. Minsky M, Papert S. 1969 *Perceptrons; an introduction to computational geometry*, p. 258. Cambridge, MA: MIT Press.
5. Rosenblatt F. 1958 The perceptron: a probabilistic model for information storage and organization in the brain. *Psychol. Rev.* **65**, 386–408. (doi:10.1037/h0042519)
6. Hopfield JJ. 1982 Neural networks and physical systems with emergent collective computational abilities. *Proc. Natl Acad. Sci. USA* **79**, 2554–2558. (doi:10.1073/pnas.79.8.2554)
7. Hopfield JJ, Tank DW. 1986 Computing with neural circuits: a model. *Science* **233**, 625–633. (doi:10.1126/science.3755256)
8. Yang CY, Bialecka-Fornal M, Weatherwax C, Larkin JW, Prindle A, Liu J, Garcia-Ojalvo J, Süel GM. 2020 Encoding membrane-potential-based memory within a microbial community. *Cell Syst.* **10**, 417–423. (doi:10.1016/j.cels.2020.04.002)
9. Martinez-Corral R, Liu J, Prindle A, Süel GM, Garcia-Ojalvo J. 2019 Metabolic basis of brain-like electrical signalling in bacterial communities. *Phil. Trans. R. Soc. B Biol. Sci.* **374**, 20180382. (doi:10.1098/rstb.2018.0382)
10. Prindle A, Liu J, Asally M, Ly S, Garcia-Ojalvo J, Süel GM. 2015 Ion channels enable electrical communication in bacterial communities. *Nature* **527**, 59–63. (doi:10.1038/nature15709)
11. Levin M. 2023 Bioelectric networks: the cognitive glue enabling evolutionary scaling from physiology to mind. *Anim. Cogn.* **26**, 1865–1891. (doi:10.1007/s10071-023-01780-3)
12. Vallverdú J *et al.* 2018 Slime mould: the fundamental mechanisms of biological cognition. *BioSystems* **165**, 57–70. (doi:10.1016/j.biosystems.2017.12.011)
13. Baluška F, Levin M. 2016 On having no head: cognition throughout biological systems. *Front. Psychol.* **7**, 902. (doi:10.3389/fpsyg.2016.00902)
14. Lyon P. 2020 Of what is ‘minimal cognition’ the half-baked version? *Adapt. Behav.* **28**, 407–424. (doi:10.1177/1059712319871360)
15. Lyon P. 2006 The biogenic approach to cognition. *Cogn. Process.* **7**, 11–29. (doi:10.1007/s10339-005-0016-8)
16. Pezzulo G, Levin M. 2015 Re-membering the body: applications of computational neuroscience to the top-down control of regeneration of limbs and other complex organs. *Integr. Biol.* **7**, 1487–1517. (doi:10.1039/c5ib00221d)
17. Levin M. 2022 Technological approach to mind everywhere: an experimentally-grounded framework for understanding diverse bodies and minds. *Front. Syst. Neurosci.* **16**, 768201. (doi:10.3389/fnsys.2022.768201)
18. Levin M. 2019 The computational boundary of a ‘self’: developmental bioelectricity drives multicellularity and scale-free cognition. *Front. Psychol.* **10**, 2688. (doi:10.3389/fpsyg.2019.02688)
19. Fields C, Levin M. 2022 Competency in navigating arbitrary spaces as an invariant for analyzing cognition in diverse embodiments. *Entropy* **24**, 819. (doi:10.3390/e24060819)

20. Chalmers D. 2017 The hard problem of consciousness. In *The Blackwell companion to consciousness* (eds S Schneider, M Velman), pp. 32–42. New York, NY: John Wiley & Sons. (doi:10.1002/9781119132363)
21. Seth AK, Bayne T. 2022 Theories of consciousness. *Nat. Rev. Neurosci.* **23**, 439–452. (doi:10.1038/s41583-022-00587-4)
22. Kuhn RL. 2024 A landscape of consciousness: toward a taxonomy of explanations and implications. *Prog. Biophys. Mol. Biol.* **190**, 28–169. (doi:10.1016/j.pbiomolbio.2023.12.003)
23. Malcolm N. 1958 I. Knowledge of other minds. *J. Philos.* **55**, 969–978. (doi:10.2307/2021905)
24. Price HH. 1938 Our evidence for the existence of other minds. *Philosophy* **13**, 425–456. (doi:10.1017/S0031819100014200)
25. Sober E. 1995 Why not solipsism? *Philos. Phenomenol. Res.* **55**, 547–566. (doi:10.2307/2108438)
26. Bayne T *et al.* 2024 Tests for consciousness in humans and beyond. *Trends Cogn. Sci.* **28**, 454–466. (doi:10.1016/j.tics.2024.01.010)
27. Povinelli DJ, Giambrone S. 1999 Inferring other minds: failure of the argument by analogy. *Philos. Top.* **27**, 167–201. (doi:10.5840/philtopics199927120)
28. Frith U. 2001 Mind blindness and the brain in autism. *Neuron* **32**, 969–979. (doi:10.1016/s0896-6273(01)00552-9)
29. Hoffman DD. 2019 *The case against reality: why evolution hid the truth from our eyes*. New York, NY: W. W. Norton & Company.
30. Hoffman DD, Singh M, Prakash C. 2015 The interface theory of perception. *Psychon. Bull. Rev.* **22**, 1480–1506. (doi:10.3758/s13423-015-0890-8)
31. Shirakawa T, Gunji YP, Miyake Y. 2011 An associative learning experiment using the plasmodium of *Physarum polycephalum*. *Nano Commun. Netw.* **2**, 99–105. (doi:10.1016/j.nancom.2011.05.002)
32. Hennessey TM, Rucker WB, McDiarmid CG. 1979 Classical-conditioning in paramecia. *Anim. Learn. Behav.* **7**, 417–423. (doi:10.3758/BF03209695)
33. Biswas S, Clawson W, Levin M. 2022 Learning in transcriptional network models: computational discovery of pathway-level memory and effective interventions. *Int. J. Mol. Sci.* **24**, 285. (doi:10.3390/ijms24010285)
34. Biswas S, Manicka S, Hoel E, Levin M. 2021 Gene regulatory networks exhibit several kinds of memory: quantification of memory in biological and random transcriptional networks. *iScience* **24**, 102131. (doi:10.1016/j.isci.2021.102131)
35. Kukushkin NV, Carney RE, Tabassum T, Carew TJ. 2024 The massed-spaced learning effect in non-neural human cells. *Nat. Commun.* **15**, 9635. (doi:10.1038/s41467-024-53922-x)
36. Gershman SJ, Balbi PE, Gallistel CR, Gunawardena J. 2021 Reconsidering the evidence for learning in single cells. *eLife* **10**, e61907. (doi:10.7554/eLife.61907)
37. Katz Y, Fontana W. 2022 Probabilistic inference with polymerizing biochemical circuits. *Entropy* **24**, 629. (doi:10.3390/e24050629)
38. Katz Y, Springer M, Fontana W. 2018 Embodying probabilistic inference in biochemical circuits. *arXiv:1806.10161*.
39. Kaygisiz K, Ulijn RV. 2024 Can molecular systems learn? *ChemSystemsChem* **7**, e202400075. (doi:10.1002/syst.202400075)
40. Levin M, Martyniuk CJ. 2018 The bioelectric code: an ancient computational medium for dynamic control of growth and form. *BioSystems* **164**, 76–93. (doi:10.1016/j.biosystems.2017.08.009)
41. Turing AM. 1952 The chemical basis of morphogenesis. *Phil. Trans. R. Soc. Lond. B Biol. Sci.* **237**, 37–72. (doi:10.1098/rstb.1952.0012)
42. Rouleau N, Levin M. 2024 Discussions of machine versus living intelligence need more clarity. *Nat. Mach. Intell.* **6**, 1424–1426. (doi:10.1038/s42256-024-00955-y)
43. Yampolskiy RV. 2014 The universe of minds. *ArXiv abs/1410.0369*.
44. Dennett DC. 1996 *Kinds of minds: toward an understanding of consciousness*, p. 184, 1st edn. New York, NY: Basic Books. (doi:10.1002/cplx.6130010605)
45. Nagel T. 1995 *Other minds: critical essays*. Oxford, UK: Oxford University Press.
46. Sloman A. 1984 The structure of the space of possible minds. In *The mind and the machine: philosophical aspects of artificial intelligence* (eds S Torrance, E Horwood), pp. 35–42. Chichester, UK: MIT.

47. Morgan CL. 1903 Other minds than ours. In *An introduction to comparative psychology* (ed. W Scott), pp. 36–59. London, UK: Walter Scott Publishing Co. (doi:10.1037/13701-003)
48. Koch C, Massimini M, Boly M, Tononi G. 2016 Neural correlates of consciousness: progress and problems. *Nat. Rev. Neurosci.* **17**, 307–321. (doi:10.1038/nrn.2016.22)
49. Di Perri C *et al.* 2016 Neural correlates of consciousness in patients who have emerged from a minimally conscious state: a cross-sectional multimodal imaging study. *Lancet Neurol.* **15**, 830–842. (doi:10.1016/S1474-4422(16)00111-3)
50. Gawryluk JR, D'Arcy RCN, Connolly JF, Weaver DF. 2010 Improving the clinical assessment of consciousness with advances in electrophysiological and neuroimaging techniques. *BMC Neurol.* **10**, 11. (doi:10.1186/1471-2377-10-11)
51. Boly M, Massimini M, Tononi G. 2009 Theoretical approaches to the diagnosis of altered states of consciousness. *Prog. Brain Res.* **177**, 383–398. (doi:10.1016/S0079-6123(09)17727-0)
52. Tononi G, Koch C. 2015 Consciousness: here, there and everywhere? *Phil. Trans. R. Soc. B Biol. Sci.* **370**, 20140167. (doi:10.1098/rstb.2014.0167)
53. Poldrack RA. 2011 Inferring mental states from neuroimaging data: from reverse inference to large-scale decoding. *Neuron* **72**, 692–697. (doi:10.1016/j.neuron.2011.11.001)
54. Sinitsyn DO *et al.* 2020 Detecting the potential for consciousness in unresponsive patients using the perturbational complexity index. *Brain Sci.* **10**, 917. (doi:10.3390/brainsci10120917)
55. Casarotto S *et al.* 2016 Stratification of unresponsive patients by an independently validated index of brain complexity. *Ann. Neurol.* **80**, 718–729. (doi:10.1002/ana.24779)
56. Spence CE, Osman M, McElligott AG. 2017 Theory of animal mind: human nature or experimental artefact? *Trends Cogn. Sci.* **21**, 333–343. (doi:10.1016/j.tics.2017.02.003)
57. Urquiza-Haas EG, Kotrschal K. 2015 The mind behind anthropomorphic thinking: attribution of mental states to other species. *Anim. Behav.* **109**, 167–176. (doi:10.1016/j.anbehav.2015.08.011)
58. Harnad S. 2016 Animal sentience: the other-minds problem. *Animal Sentience* **1**, 1. (doi:10.51291/2377-7478.1065)
59. DeGrazia D. 1997 Great apes, dolphins, and the concept of personhood + moral philosophy, categories. *South. J. Philos.* **35**, 301–320. (doi:10.1111/j.2041-6962.1997.tb00839.x)
60. Staker A. 2017 Should chimpanzees have standing? The case for pursuing legal personhood for non-human animals. *Transnatl. Environ. Law* **6**, 485–507. (doi:10.1017/S204710251700019X)
61. Bar-On YM, Phillips R, Milo R. 2018 The biomass distribution on Earth. *Proc. Natl Acad. Sci. USA* **115**, 6506–6511. (doi:10.1073/pnas.1711842115)
62. Jennings HS. 1906 *Behavior of the lower organisms*, pp. 14–366, vol. xiv. New York, NY: The Columbia University Press, The Macmillan Company, Agents.
63. Baluška F, Reber AS, Miller WB. 2022 Cellular sentience as the primary source of biological order and evolution. *BioSystems* **218**, 104694. (doi:10.1016/j.biosystems.2022.104694)
64. Baluška F, Miller WB, Reber AS. 2022 Cellular and evolutionary perspectives on organismal cognition: from unicellular to multicellular organisms. *Biol. J. Linn. Soc.* **139**, 503–513. (doi:10.1093/biolinnean/blac005)
65. Reber AS, Baluška F. 2021 Cognition in some surprising places. *Biochem. Biophys. Res. Commun.* **564**, 150–157. (doi:10.1016/j.bbrc.2020.08.115)
66. DeMarse TB, Wagenaar DA, Blau AW, Potter SM. 2000 Interfacing neuronal cultures to a computer-generated virtual world. In *7th Joint Symposium on Neural Computation*, Los Angeles, CA, pp. 36–42. Institute for Neural Computation.
67. Potter SM, Fraser SE, Pine J. 1997 Animat in a Petri dish: cultured neural networks for studying neural computation. In *4th Joint Symposium on Neural Computation*, San Diego, CA, pp. 167–174. Institute for Neural Computation.
68. Demarse TB, Wagenaar DA, Blau AW, Potter SM. 2001 The neurally controlled animat: biological brains acting with simulated bodies. *Auton. Robots* **11**, 305–310. (doi:10.1023/a:1012407611130)
69. Wagenaar DA, Madhavan R, Pine J, Potter SM. 2005 Controlling bursting in cortical cultures with closed-loop multi-electrode stimulation. *J. Neurosci.* **25**, 680–688. (doi:10.1523/JNEUROSCI.4209-04.2005)

70. Madhavan R, Wagenaar DA, Potter SM. 2003 Multisite stimulation quiets bursts and enhances plasticity in cultured networks. In *33rd Society for Neuroscience Annual Meeting*, New Orleans, LA, pp. 45–56. Society for Neuroscience.
71. Potter AM *et al.* 2004 Hybrot: hybrids of living neurons and robots for studying neural computation. In *Brain inspired cognitive systems* (eds AMG Peebles, RP Cooper), pp. 1–6. Stirling, UK: World Scientific.
72. Lobov SA, Mikhaylov AN, Shamshin M, Makarov VA, Kazantsev VB. 2020 Spatial properties of STDP in a self-learning spiking neural network enable controlling a mobile robot. *Front. Neurosci.* **14**, 88. (doi:10.3389/fnins.2020.00088)
73. Ebrahimkhani MR, Levin M. 2021 Synthetic living machines: a new window on life. *iScience* **24**, 102505. (doi:10.1016/j.isci.2021.102505)
74. Kamm RD *et al.* 2018 Perspective: the promise of multi-cellular engineered living systems. *APL Bioeng.* **2**, 040901. (doi:10.1063/1.5038337)
75. Blackiston D, Kriegman S, Bongard J, Levin M. 2023 Biological robots: perspectives on an emerging interdisciplinary field. *Soft Robot.* **10**, 674–686. (doi:10.1089/soro.2022.0142)
76. Gumuskaya G, Srivastava P, Cooper BG, Lesser H, Semegran B, Garnier S, Levin M. 2023 Motile living biobots self-construct from adult human somatic progenitor seed cells. *Adv. Sci.* **11**, e2303575. (doi:10.1002/advs.202303575)
77. Clawson WP, Levin M. 2022 Endless forms most beautiful 2.0: teleonomy and the bioengineering of chimaeric and synthetic organisms. *Biol. J. Linn. Soc.* **139**, 457–486. (doi:10.1093/biolinnean/blac073)
78. Pio-Lopez L. 2021 The rise of the biocyborg: synthetic biology, artificial chimerism and human enhancement. *New Genet. Soc.* **40**, 599–619. (doi:10.1080/14636778.2021.2007064)
79. Orive G, Taebnia N, Dolatshahi-Pirouz A. 2020 A new era for cyborg science is emerging: the promise of cyborganic beings. *Adv. Healthc. Mater.* **9**, e1901023. (doi:10.1002/adhm.201901023)
80. Li Q, Nan K, Le Floch P, Lin Z, Sheng H, Blum TS, Liu J. 2019 Cyborg organoids: implantation of nanoelectronics via organogenesis for tissue-wide electrophysiology. *Nano Lett.* **19**, 5781–5789. (doi:10.1021/acs.nanolett.9b02512)
81. Mehrali M *et al.* 2018 Blending electronics with the human body: a pathway toward a cybernetic future. *Adv. Sci.* **5**, 1700931. (doi:10.1002/advs.201700931)
82. Ding S, O'Banion CP, Welfare JG, Lawrence DS. 2018 Cellular cyborgs: on the precipice of a drug delivery revolution. *Cell Chem. Biol.* **25**, 648–658. (doi:10.1016/j.chembiol.2018.03.003)
83. Aaser P *et al.* 2017 Towards making a cyborg: a closed-loop reservoir-neuro system. In *ECAL 2017: the 14th European Conf. on Artificial Life*, pp. 430–437. Lyon, France: MIT Press. (doi:10.7551/ecal_a_072)
84. Stern M, Pinson MB, Murugan A. 2020 Continual learning of multiple memories in mechanical networks. *Phys. Rev. X* **10**, 031044. (doi:10.1103/PhysRevX.10.031044)
85. Stern M, Arinze C, Perez L, Palmer SE, Murugan A. 2020 Supervised learning through physical changes in a mechanical system. *Proc. Natl Acad. Sci. USA* **117**, 14843–14850. (doi:10.1073/pnas.2000807117)
86. Ozkan-Aydin Y, Goldman DI, Bhamla MS. 2021 Collective dynamics in entangled worm and robot blobs. *Proc. Natl Acad. Sci. USA* **118**, e2010542118. (doi:10.1073/pnas.2010542118)
87. Bernheim - Groswasser A, Gov NS, Safran SA, Tzlil S. 2018 Living matter: mesoscopic active materials. *Adv. Mater.* **30**, e1707028. (doi:10.1002/adma.201707028)
88. Čejková J, Banno T, Hanczyc MM, Štěpánek F. 2017 Droplets as liquid robots. *Artif. Life* **23**, 528–549. (doi:10.1162/ARTL_a_00243)
89. Rouleau N, Levin M. 2023 The multiple realizability of sentience in living systems and beyond. *eNeuro* **10**, ENEURO.0375-23.2023. (doi:10.1523/ENEURO.0375-23.2023)
90. Rosenblueth A, Wiener N, Bigelow J. 1943 Behavior, purpose and teleology. *Philos. Sci.* **10**, 18–24. (doi:10.1086/286788)
91. Levin M. 2024 The multiscale wisdom of the body: collective intelligence as a tractable interface for next-generation biomedicine. *BioEssays* **47**, e202400196. (doi:10.1002/bies.202400196)
92. Cho ES, Han S, Kim G, Eom M, Lee KH, Kim CH, Yoon YG. 2023 In vivo whole-brain imaging of zebrafish larvae using three-dimensional fluorescence microscopy. *J. Vis. Exp.* **194**, 65218. (doi:10.3791/65218)

93. Fields C, Bischof J, Levin M. 2020 Morphological coordination: a common ancestral function unifying neural and non-neural signaling. *Physiology* **35**, 16–30. (doi:10.1152/physiol.00027.2019)
94. Koshland DE. 1983 The bacterium as a model neuron. *Trends Neurosci.* **6**, 133–137. (doi:10.1016/0166-2236(83)90066-8)
95. Koshland DE. 1980 Bacterial chemotaxis in relation to neurobiology. *Annu. Rev. Neurosci.* **3**, 43–75. (doi:10.1146/annurev.ne.03.030180.000355)
96. McMillen P, Oudin MJ, Levin M, Payne SL. 2021 Beyond neurons: long distance communication in development and cancer. *Front. Cell Dev. Biol.* **9**, 739024. (doi:10.3389/fcell.2021.739024)
97. Blackiston D, Adams DS, Lemire JM, Lobikin M, Levin M. 2011 Transmembrane potential of GlyCl-expressing instructor cells induces a neoplastic-like conversion of melanocytes via a serotonergic pathway. *Dis. Model. Mech.* **4**, 67–85. (doi:10.1242/dmm.005561)
98. Hunt T, Jones M, McFadden J, Delorme A, Hales CG, Ericson M, Schooler J. 2023 Editorial: electromagnetic field theories of consciousness: opportunities and obstacles. *Front. Hum. Neurosci.* **17**, 1342634. (doi:10.3389/fnhum.2023.1342634)
99. Kitchener PD, Hales CG. 2022 What neuroscientists think, and don't think, about consciousness. *Front. Hum. Neurosci.* **16**, 767612. (doi:10.3389/fnhum.2022.767612)
100. Hales CG, Ericson M. 2022 Electromagnetism's bridge across the explanatory gap: how a neuroscience/physics collaboration delivers explanation into all theories of consciousness. *Front. Hum. Neurosci.* **16**, 836046. (doi:10.3389/fnhum.2022.836046)
101. Hales CG. 2014 The origins of the brain's endogenous electromagnetic field and its relationship to provision of consciousness. *J. Integr. Neurosci.* **13**, 313–361. (doi:10.1142/S0219635214400056)
102. Pinotsis DA, Miller EK. 2023 In vivo ephaptic coupling allows memory network formation. *Cereb. Cortex* **33**, 9877–9895. (doi:10.1093/cercor/bhad251)
103. Pinotsis DA, Fridman G, Miller EK. 2023 Cytoelectric coupling: electric fields sculpt neural activity and 'tune' the brain's infrastructure. *Prog. Neurobiol.* **226**, 102465. (doi:10.1016/j.pneurobio.2023.102465)
104. Pinotsis DA, Miller EK. 2022 Beyond dimension reduction: stable electric fields emerge from and allow representational drift. *Neuroimage* **253**, 119058. (doi:10.1016/j.neuroimage.2022.119058)
105. Draper MH, Weidmann S. 1951 Cardiac resting and action potentials recorded with an intracellular electrode. *J. Physiol.* **115**, 74–94. (doi:10.1113/jphysiol.1951.sp004653)
106. Chakravarthy SV, Ghosh J. 1997 On Hebbian-like adaptation in heart muscle: a proposal for 'cardiac memory'. *Biol. Cybern.* **76**, 207–215. (doi:10.1007/s004220050333)
107. Zoghi M. 2004 Cardiac memory: do the heart and the brain remember the same? *J. Interv. Card. Electrophysiol.* **11**, 177–182. (doi:10.1023/B:JICE.0000048567.18088.a2)
108. Kurtz J. 2005 Specific memory within innate immune systems. *Trends Immunol.* **26**, 186–192. (doi:10.1016/j.it.2005.02.001)
109. Banskota S, Ghia JE, Khan WI. 2019 Serotonin in the gut: blessing or a curse. *Biochimie* **161**, 56–64. (doi:10.1016/j.biochi.2018.06.008)
110. Levin M, Buznikov GA, Lauder JM. 2006 Of minds and embryos: left-right asymmetry and the serotonergic controls of pre-neural morphogenesis. *Dev. Neurosci.* **28**, 171–185. (doi:10.1159/000091915)
111. Segundo-Ortín M, Calvo P. 2023 Plant sentience? Between romanticism and denial: science. *Animal Sentience* **8**, 1. (doi:10.51291/2377-7478.1772)
112. Miguel-Tomé S, Llinás RR. 2021 Broadening the definition of a nervous system to better understand the evolution of plants and animals. *Plant Signal. Behav.* **16**, 1927562. (doi:10.1080/15592324.2021.1927562)
113. Alipour A, Dorvash M, Yeganeh Y, Hatam G. 2018 Paramecium learning: new insights. *J. Protozool. Res.* **28**, 22–32. (doi:10.32268/jprotozoolres.28.1-2_22)
114. Boisseau RP, Vogel D, Dussutour A. 2016 Habituation in non-neural organisms: evidence from slime moulds. *Proc. R. Soc. B* **283**, 20160446. (doi:10.1098/rspb.2016.0446)
115. Boussard A, Delescluse J, Pérez-Escudero A, Dussutour A. 2019 Memory inception and preservation in slime moulds: the quest for a common mechanism. *Phil. Trans. R. Soc. B Biol. Sci.* **374**, 20180368. (doi:10.1098/rstb.2018.0368)

116. Murugan NJ *et al.* 2021 Mechanosensation mediates long - range spatial decision - making in an a neural organism. *Adv. Mater.* **33**, e2008161. (doi:[10.1002/adma.202008161](https://doi.org/10.1002/adma.202008161))
117. Bonzanni M, Rouleau N, Levin M, Kaplan DL. 2019 On the generalization of habituation: how discrete biological systems respond to repetitive stimuli: a novel model of habituation that is independent of any biological system. *BioEssays* **41**, e1900028. (doi:[10.1002/bies.201900028](https://doi.org/10.1002/bies.201900028))
118. Bonzanni M, Rouleau N, Levin M, Kaplan DL. 2020 Optogenetically induced cellular habituation in non-neuronal cells. *PLoS One* **15**, e0227230. (doi:[10.1371/journal.pone.0227230](https://doi.org/10.1371/journal.pone.0227230))
119. Lyon P. 2015 The cognitive cell: bacterial behavior reconsidered. *Front. Microbiol.* **6**, 264. (doi:[10.3389/fmicb.2015.00264](https://doi.org/10.3389/fmicb.2015.00264))
120. Keijzer F, van Duijn M, Lyon P. 2013 What nervous systems do: early evolution, input-output, and the skin brain thesis. *Adapt. Behav.* **21**, 67–85. (doi:[10.1177/1059712312465330](https://doi.org/10.1177/1059712312465330))
121. Calvo P, Baluška F. 2015 Conditions for minimal intelligence across eukaryota: a cognitive science perspective. *Front. Psychol.* **6**, 1329. (doi:[10.3389/fpsyg.2015.01329](https://doi.org/10.3389/fpsyg.2015.01329))
122. Grémiaux A, Yokawa K, Mancuso S, Baluška F. 2014 Plant anesthesia supports similarities between animals and plants: Claude Bernard's forgotten studies. *Plant Signal. Behav.* **9**, e27886. (doi:[10.4161/psb.27886](https://doi.org/10.4161/psb.27886))
123. Trewavas AJ, Baluška F. 2011 The ubiquity of consciousness. *EMBO Rep.* **12**, 1221–1225. (doi:[10.1038/embor.2011.218](https://doi.org/10.1038/embor.2011.218))
124. Aur D. 2012 From neuroelectrodynamics to thinking machines. *Cognit. Comput.* **4**, 4–12. (doi:[10.1007/s12559-011-9106-3](https://doi.org/10.1007/s12559-011-9106-3))
125. Scott HL *et al.* 2022 Evidence for long-term potentiation in phospholipid membranes. *Proc. Natl Acad. Sci. USA* **119**, e2212195119. (doi:[10.1073/pnas.2212195119](https://doi.org/10.1073/pnas.2212195119))
126. Zheng JM, Chin WC, Khijniak E, Khijniak E, Pollack GH. 2006 Surfaces and interfacial water: evidence that hydrophilic surfaces have long-range impact. *Adv. Colloid Interface Sci.* **127**, 19–27. (doi:[10.1016/j.cis.2006.07.002](https://doi.org/10.1016/j.cis.2006.07.002))
127. Craddock TJ, Hameroff SR, Ayoub AT, Klobukowski M, Tuszynski JA. 2015 Anesthetics act in quantum channels in brain microtubules to prevent consciousness. *Curr. Top. Med. Chem.* **15**, 523–533. (doi:[10.2174/1568026615666150225104543](https://doi.org/10.2174/1568026615666150225104543))
128. Hameroff S. 2014 Quantum walks in brain microtubules-a biomolecular basis for quantum cognition? *Top. Cogn. Sci.* **6**, 91–97. (doi:[10.1111/tops.12068](https://doi.org/10.1111/tops.12068))
129. Hameroff SR, Craddock TJ, Tuszynski JA. 2010 'Memory bytes' - molecular match for CaMKII phosphorylation encoding of microtubule lattices. *J. Integr. Neurosci.* **9**, 253–267. (doi:[10.1142/s0219635210002482](https://doi.org/10.1142/s0219635210002482))
130. Hameroff S, Nip A, Porter M, Tuszynski J. 2002 Conduction pathways in microtubules, biological quantum computation, and consciousness. *BioSystems* **64**, 149–168. (doi:[10.1016/s0303-2647\(01\)00183-6](https://doi.org/10.1016/s0303-2647(01)00183-6))
131. Dimonte A, Adamatzky A, Erokhin V, Levin M. 2016 On chirality of slime mould. *BioSystems* **140**, 23–27. (doi:[10.1016/j.biosystems.2015.12.008](https://doi.org/10.1016/j.biosystems.2015.12.008))
132. Lewis OL, Zhang S, Guy RD, del Álamo JC. 2015 Coordination of contractility, adhesion and flow in migrating *Physarum amoebae*. *J. R. Soc. Interface* **12**, 20141359. (doi:[10.1098/rsif.2014.1359](https://doi.org/10.1098/rsif.2014.1359))
133. Wright M *et al.* 1988 Microtubule cytoskeleton and morphogenesis in the amoebae of the myxomycete *Physarum polycephalum*. *Biol. Cell* **63**, 239–248. (doi:[10.1016/0248-4900\(88\)90061-5](https://doi.org/10.1016/0248-4900(88)90061-5))
134. McMillen P, Levin M. 2024 Collective intelligence: a unifying concept for integrating biology across scales and substrates. *Commun. Biol.* **7**, 378. (doi:[10.1038/s42003-024-06037-4](https://doi.org/10.1038/s42003-024-06037-4))
135. Kim T, Kao MT, Hasselbrink EF, Meyhöfer E. 2007 Active alignment of microtubules with electric fields. *Nano Lett.* **7**, 211–217. (doi:[10.1021/nl061474k](https://doi.org/10.1021/nl061474k))
136. Stracke R, Böhm KJ, Wollweber L, Tuszynski JA, Unger E. 2002 Analysis of the migration behaviour of single microtubules in electric fields. *Biochem. Biophys. Res. Commun.* **293**, 602–609. (doi:[10.1016/S0006-291X\(02\)00251-6](https://doi.org/10.1016/S0006-291X(02)00251-6))
137. Liao M, Liang X, Howard J. 2021 The narrowing of dendrite branches across nodes follows a well-defined scaling law. *Proc. Natl Acad. Sci. USA* **118**, e2022395118. (doi:[10.1073/pnas.2022395118](https://doi.org/10.1073/pnas.2022395118))

138. Brummer AB *et al.* 2021 Branching principles of animal and plant networks identified by combining extensive data, machine learning and modelling. *J. R. Soc. Interface* **18**, 20200624. (doi:10.1098/rsif.2020.0624)
139. Cuntz H, Mathy A, Häusser M. 2012 A scaling law derived from optimal dendritic wiring. *Proc. Natl Acad. Sci. USA* **109**, 11014–11018. (doi:10.1073/pnas.1200430109)
140. Cragg BG, Temperley HNV. 1955 Memory: the analogy with ferromagnetic hysteresis. *Brain* **78**, 304–316. (doi:10.1093/brain/78.2.304)
141. Loeffler A, Diaz-Alvarez A, Zhu R, Ganesh N, Shine JM, Nakayama T, Kuncic Z. 2023 Neuromorphic learning, working memory, and metaplasticity in nanowire networks. *Sci. Adv.* **9**, eadg3289. (doi:10.1126/sciadv.adg3289)
142. Persinger MA. 2012 Brain electromagnetic activity and lightning: potentially congruent scale-invariant quantitative properties. *Front. Integr. Neurosci.* **6**, 19. (doi:10.3389/fnint.2012.00019)
143. Park HJ, Friston K. 2013 Structural and functional brain networks: from connections to cognition. *Science* **342**, 1238411. (doi:10.1126/science.1238411)
144. John ER. 2002 The neurophysics of consciousness. *Brain Res. Brain Res. Rev.* **39**, 1–28. (doi:10.1016/s0165-0173(02)00142-x)
145. Koch C. 2018 What is consciousness? *Sci. Am.* **318**, 60–64. (doi:10.1038/scientificamerican0618-60)
146. Rouleau N, Levin M. 2023 Multiple ways to implement and infer sentience. *Animal Sentience* **8**, 30. (doi:10.51291/2377-7478.1817)
147. Friston K. 2018 Am I self-conscious? (Or does self-organization entail self-consciousness?). *Front. Psychol.* **9**, 579. (doi:10.3389/fpsyg.2018.00579)
148. Solms M. 2018 The hard problem of consciousness and the free energy principle. *Front. Psychol.* **9**, 2714. (doi:10.3389/fpsyg.2018.02714)
149. Graziano MSA. 2017 The attention schema theory: a foundation for engineering artificial consciousness. *Front. Robot. AI* **4**, 60. (doi:10.3389/frobt.2017.00060)
150. Jackendoff R. 1987 *Consciousness and the computational mind*. Cambridge, MA: MIT Press.
151. Prinz J. 2012 *The conscious brain: how attention engenders experience*. Oxford, UK: Oxford University Press.
152. Seth AK. 2021 *Being you: a new science of consciousness*. London, UK: Faber & Faber.
153. Seth AK. 2015 The cybernetic Bayesian brain. In *Open mind* (eds TK Metzinger, JM Windt). Frankfurt am Main, Germany: MIND Group.
154. Seth AK, Tsakiris M. 2018 Being a beast machine: the somatic basis of selfhood. *Trends Cogn. Sci.* **22**, 969–981. (doi:10.1016/j.tics.2018.08.008)
155. Aru J, Suzuki M, Larkum ME. 2020 Cellular mechanisms of conscious processing. *Trends Cogn. Sci.* **24**, 814–825. (doi:10.1016/j.tics.2020.07.006)
156. Tononi G, Edelman GM. 1998 Consciousness and complexity. *Science* **282**, 1846–1851. (doi:10.1126/science.282.5395.1846)
157. McFadden J. 2020 Integrating information in the brain's EM field: the cemi field theory of consciousness. *Neurosci. Conscious.* **2020**, niaa016. (doi:10.1093/nc/niaa016)
158. Baars BJ. 1988 *A cognitive theory of consciousness*. Cambridge, UK: Cambridge University Press.
159. Dehaene S, Changeux JP. 2011 Experimental and theoretical approaches to conscious processing. *Neuron* **70**, 200–227. (doi:10.1016/j.neuron.2011.03.018)
160. Mashour GA, Roelfsema P, Changeux JP, Dehaene S. 2020 Conscious processing and the global neuronal workspace hypothesis. *Neuron* **105**, 776–798. (doi:10.1016/j.neuron.2020.01.026)
161. Rosenthal D. 2005 *Consciousness and mind*. Oxford, UK: Clarendon Press.
162. Brown R, Lau H, LeDoux JE. 2019 Understanding the higher-order approach to consciousness. *Trends Cogn. Sci.* **23**, 754–768. (doi:10.1016/j.tics.2019.06.009)
163. Chang AYC, Biehler M, Yu Y, Kanai R. 2020 Information closure theory of consciousness. *Front. Psychol.* **11**, 1504. (doi:10.3389/fpsyg.2020.01504)
164. Tononi G. 2008 Consciousness as integrated information: a provisional manifesto. *Biol. Bull.* **215**, 216–242. (doi:10.2307/25470707)

165. Tononi G, Boly M, Massimini M, Koch C. 2016 Integrated information theory: from consciousness to its physical substrate. *Nat. Rev. Neurosci.* **17**, 450–461. (doi:10.1038/nrn.2016.44)
166. Oizumi M, Albantakis L, Tononi G. 2014 From the phenomenology to the mechanisms of consciousness: integrated information theory 3.0. *PLoS Comput. Biol.* **10**, e1003588. (doi:10.1371/journal.pcbi.1003588)
167. Lamme VAF. 2006 Towards a true neural stance on consciousness. *Trends Cogn. Sci.* **10**, 494–501. (doi:10.1016/j.tics.2006.09.001)
168. Lamme VAF. 2010 How neuroscience will change our view on consciousness. *Cogn. Neurosci.* **1**, 204–220. (doi:10.1080/17588921003731586)
169. Dennett DC. 1991 *Consciousness explained*. Boston, MA; New York, NY; London, UK: Little, Brown.
170. Edelman GM. 1987 *Neural darwinism: the theory of neuronal group selection*. New York, NY: Basic Books Inc.
171. Edelman GM. 1989 *The remembered present: a biological theory of consciousness*. New York, NY: Basic Books Inc.
172. Park HD, Tallon-Baudry C. 2014 The neural subjective frame: from bodily signals to perceptual consciousness. *Phil. Trans. R. Soc. B Biol. Sci.* **369**, 20130208. (doi:10.1098/rstb.2013.0208)
173. Pennartz CMA. 2018 Consciousness, representation, action: the importance of being goal-directed. *Trends Cogn. Sci.* **22**, 137–153. (doi:10.1016/j.tics.2017.10.006)
174. Hameroff S, Penrose R. 2014 Consciousness in the universe: a review of the ‘Orch OR’ theory. *Phys. Life Rev.* **11**, 39–78. (doi:10.1016/j.plprev.2013.08.002)
175. Hohwy J, Seth A. 2020 Predictive processing as a systematic basis for identifying the neural correlates of consciousness. *PhiMiSci* **1**, 3. (doi:10.33735/phemisci.2020.II.64)
176. Clark A. 2013 Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behav. Brain Sci.* **36**, 181–204. (doi:10.1017/S0140525X12000477)
177. Hohwy J. 2013 *The predictive mind*. Oxford, UK: Oxford University Press.
178. Marvan T, Havlík M. 2021 Is predictive processing a theory of perceptual consciousness? *New Ideas Psychol.* **61**, 100837. (doi:10.1016/j.newideapsych.2020.100837)
179. O’Regan JK, Noë A. 2001 A sensorimotor account of vision and visual consciousness. *Behav. Brain Sci.* **24**, 939–973. (doi:10.1017/s0140525x01000115)
180. Damasio A. 2010 *Self comes to mind: constructing the conscious brain*. New York, NY: Pantheon.
181. Damasio A. 2000 *The feeling of what happens: body and emotion in the making of consciousness*. New York, NY: HarperCollins.
182. Cleeremans A, Achoui D, Beauny A, Keuninckx L, Martin JR, Muñoz-Moldes S, Vuillaume L, de Heering A. 2020 Learning to be conscious. *Trends Cogn. Sci.* **24**, 112–123. (doi:10.1016/j.tics.2019.11.011)
183. Ginsburg S, Jablonka E. 2019 *The evolution of the sensitive soul: learning and the origins of consciousness*. Cambridge, MA: MIT Press.
184. Birch J, Ginsburg S, Jablonka E. 2020 Unlimited associative learning and the origins of consciousness: a primer and some predictions. *Biol. Philos.* **35**, 56. (doi:10.1007/s10539-020-09772-0)
185. Vandenberg LN, Adams DS, Levin M. 2012 Normalized shape and location of perturbed craniofacial structures in the *Xenopus tadpole* reveal an innate ability to achieve correct morphology. *Dev. Dyn.* **241**, 863–878. (doi:10.1002/dvdy.23770)
186. Vandenberg LN, Morrie RD, Adams DS. 2011 V-ATPase-dependent ectodermal voltage and pH regionalization are required for craniofacial morphogenesis. *Dev. Dyn.* **240**, 1889–1904. (doi:10.1002/dvdy.22685)
187. Pezzulo G, Levin M. 2016 Top-down models in biology: explanation and control of complex living systems above the molecular level. *J. R. Soc. Interface* **13**, 20160555. (doi:10.1098/rsif.2016.0555)
188. Blackiston DJ, Levin M. 2013 Ectopic eyes outside the head in *Xenopus* tadpoles provide sensory data for light-mediated learning. *J. Exp. Biol.* **216**, 1031–1040. (doi:10.1242/jeb.074963)

189. Fankhauser G. 1945 Maintenance of normal structure in heteroploid salamander larvae, through compensation of changes in cell size by adjustment of cell number and cell shape. *J. Exp. Zool.* **100**, 445–455. (doi:10.1002/jez.1401000310)
190. Levin M. 2023 Darwin's agential materials: evolutionary implications of multiscale competency in developmental biology. *Cell. Mol. Life Sci.* **80**, 142. (doi:10.1007/s00018-023-04790-z)
191. Levin M. 2023 Collective intelligence of morphogenesis as a teleonomic process. In *Evolution 'on purpose': teleonomy in living systems* (eds PA Corning, SA Kauffman, D Noble, JA Shapiro, RI Vane-Wright, A Pross), pp. 175–198. Cambridge, UK: The MIT Press. (doi:10.7551/mitpress/14642.003.0013)
192. Lyon P, Keijzer F, Arendt D, Levin M. 2021 Reframing cognition: getting down to biological basics. *Phil. Trans. R. Soc. B Biol. Sci.* **376**, 20190750. (doi:10.1098/rstb.2019.0750)
193. Levin M, Keijzer F, Lyon P, Arendt D. 2021 Uncovering cognitive similarities and differences, conservation and innovation. *Phil. Trans. R. Soc. B Biol. Sci.* **376**, 20200458. (doi:10.1098/rstb.2020.0458)
194. Pfeifer R, Bongard J, Grand S. 2007 *How the body shapes the way we think: a new view of intelligence*, pp. xxiv–394. Cambridge, MA: MIT Press. (doi:10.7551/mitpress/3585.001.0001)
195. Pfeifer R, Iida F, Lungarella M. 2014 Cognition from the bottom up: on biological inspiration, body morphology, and soft materials. *Trends Cogn. Sci.* **18**, 404–413. (doi:10.1016/j.tics.2014.04.004)
196. Pfeifer R, Gómez G. 2009 Morphological computation – connecting brain, body, and environment. In *Creating brain-like intelligence: from basic principles to complex intelligent systems* (eds B Goertzel, N Geisweiller), pp. 66–83, vol. **5436**. Springer. (doi:10.1007/978-3-642-00616-6_5)
197. Mathews J, Chang AJ, Devlin L, Levin M. 2023 Cellular signaling pathways as plastic, proto-cognitive systems: implications for biomedicine. *Patterns* **4**, 100737. (doi:10.1016/j.patter.2023.100737)
198. Lagasse E, Levin M. 2023 Future medicine: from molecular pathways to the collective intelligence of the body. *Trends Mol. Med.* **29**, 687–710. (doi:10.1016/j.molmed.2023.06.007)
199. Csermely P, Kunsic N, Mendik P, Kerestély M, Faragó T, Veres DV, Tompa P. 2020 Learning of signaling networks: molecular mechanisms. *Trends Biochem. Sci.* **45**, 284–294. (doi:10.1016/j.tibs.2019.12.005)
200. Gyurkó DM, Veres DV, Módos D, Lenti K, Korcsmáros T, Csermely P. 2013 Adaptation and learning of molecular networks as a description of cancer development at the systems-level: potential use in anti-cancer therapies. *Semin. Cancer Biol.* **23**, 262–269. (doi:10.1016/j.semcancer.2013.06.005)
201. Tononi G. 2012 Integrated information theory of consciousness: an updated account. *Arch. Ital. Biol.* **150**, 56–90. (doi:10.4449/aib.v149i5.1388)
202. Albantakis L *et al.* 2023 Integrated information theory (IIT) 4.0: formulating the properties of phenomenal existence in physical terms. *PLoS Comput. Biol.* **19**, e1011465. (doi:10.1371/journal.pcbi.1011465)
203. McMillen P, Walker SI, Levin M. 2022 Information theory as an experimental tool for integrating disparate biophysical signaling modules. *Int. J. Mol. Sci.* **23**, 9580. (doi:10.3390/ijms23179580)
204. Blackiston D *et al.* 2024 Revealing non-trivial information structures in aneural biological tissues via functional connectivity. *bioRxiv*, 2024.2005.2009.593467. (doi:10.1101/2024.05.09.593467)
205. Varley TF, Pai VP, Grasso C, Lunshof J, Levin M, Bongard J. 2024 Identification of brain-like complex information architectures in embryonic tissue of *Xenopus laevis*. *bioRxiv*. (doi:10.1101/2024.12.05.627037)
206. Pigozzi F, Goldstein A, Levin M. 2024 Associative conditioning in gene regulatory network models increases integrative causal emergence. *OSF Preprints*. (doi:10.31219/osf.io/2bc4n)
207. McMillen P, Levin M. 2024 Optical estimation of bioelectric patterns in living embryos. *Methods Mol. Biol.* **2745**, 91–102. (doi:10.1007/978-1-0716-3577-3_6)
208. Hausmann D *et al.* 2023 Autonomous rhythmic activity in glioma networks drives brain tumour growth. *Nature* **613**, 179–186. (doi:10.1038/s41586-022-05520-4)

209. Quicke P, Sun Y, Arias-Garcia M, Beykou M, Acker CD, Djamgoz MBA, Bakal C, Foust AJ. 2022 Voltage imaging reveals the dynamic electrical signatures of human breast cancer cells. *Commun. Biol.* **5**, 1178. (doi:[10.1038/s42003-022-04077-2](https://doi.org/10.1038/s42003-022-04077-2))
210. Kamm RD, Bashir R. 2014 Creating living cellular machines. *Ann. Biomed. Eng.* **42**, 445–459. (doi:[10.1007/s10439-013-0902-7](https://doi.org/10.1007/s10439-013-0902-7))
211. Ebrahimkhani MR, Ebisuya M. 2019 Synthetic developmental biology: build and control multicellular systems. *Curr. Opin. Chem. Biol.* **52**, 9–15. (doi:[10.1016/j.cbpa.2019.04.006](https://doi.org/10.1016/j.cbpa.2019.04.006))
212. Kitamura T, Tahara T, Asami KI. 2000 How can a robot have consciousness? *Adv. Robot.* **14**, 263–275. (doi:[10.1163/156855300741573](https://doi.org/10.1163/156855300741573))
213. Singh S, Chaudhary D, Gupta AD, Prakash Lohani B, Kushwaha PK, Bibhu V. 2022 Artificial intelligence, cognitive robotics and nature of consciousness. In *2022 3rd Int. Conf. on Intelligent Engineering and Management (ICIEM)*, pp. 447–454. London, UK: IEEE. (doi:[10.1109/ICIEM54221.2022.9853081](https://doi.org/10.1109/ICIEM54221.2022.9853081))
214. Cardon A. 2006 Artificial consciousness, artificial emotions, and autonomous robots. *Cogn. Process.* **7**, 245–267. (doi:[10.1007/s10339-006-0154-7](https://doi.org/10.1007/s10339-006-0154-7))
215. Rouleau N. 2022 Embodied 3D neural tissue cultures for cognitive research. In *Brain and cognitive intelligence control in robotics* (ed. B Wei), pp. 81–101. Boca Raton, FL: CRC Press. (doi:[10.1201/9781003050315-5](https://doi.org/10.1201/9781003050315-5))
216. Rouleau N, Murugan NJ, Kaplan DL. 2021 Toward studying cognition in a dish. *Trends Cogn. Sci.* **25**, 294–304. (doi:[10.1016/j.tics.2021.01.005](https://doi.org/10.1016/j.tics.2021.01.005))
217. Powers WT. 1973 *Behavior: the control of perception*, p. 296, vol. xi. Chicago, IL: Aldine Pub. Co.
218. Friston K, Samothrakis S, Montague R. 2012 Active inference and agency: optimal control without cost functions. *Biol. Cybern.* **106**, 523–541. (doi:[10.1007/s00422-012-0512-8](https://doi.org/10.1007/s00422-012-0512-8))
219. Pezzulo G, Rigoli F, Friston KJ. 2018 Hierarchical active inference: a theory of motivated control. *Trends Cogn. Sci.* **22**, 294–306. (doi:[10.1016/j.tics.2018.01.009](https://doi.org/10.1016/j.tics.2018.01.009))
220. Kirchhoff M, Parr T, Palacios E, Friston K, Kiverstein J. 2018 The Markov blankets of life: autonomy, active inference and the free energy principle. *J. R. Soc. Interface* **15**, 20170792. (doi:[10.1098/rsif.2017.0792](https://doi.org/10.1098/rsif.2017.0792))
221. Maciejewicz B. 2022 Neuroscience of consciousness in the locked-in syndrome: prognostic and diagnostic review. *Ibrain* **8**, 476–480. (doi:[10.1002/ibra.12077](https://doi.org/10.1002/ibra.12077))
222. Darwin C. 1859 *On the origin of species by means of natural selection, or the preservation of favoured races in the struggle for life*, pp. 1–502, vol. ix. London, UK: John Murray.
223. de Haan EHF *et al.* 2020 Split-brain: what we know now and why this is important for understanding consciousness. *Neuropsychol. Rev.* **30**, 224–233. (doi:[10.1007/s11065-020-09439-3](https://doi.org/10.1007/s11065-020-09439-3))